

# File Management

## File

- File is collection of data/information on storage device.
  - File = Contents (Data) + Information (Metadata)
  - The data is stored in zero or more Data blocks (in FS), while metadata is stored in the FCB (in filesystem).
- FCB is called as "inode" on UNIX/Linux. It contains
  - type: UNIX/Linux has 7 types of files
    - -: regular, d: directory, l: symbolic link, p: pipe, s: socket, c: char device, b: block device
  - size: number of bytes
  - links: number of hard links
  - mode (permissions): (u) rwx, (g) rwx, (o) rwx
  - user & group
  - time-stamps: modification, creation, access.
  - info about data blocks
- terminal> ls -l
  - type, mode, links, user, group, size, timestamp, name.
- terminal> stat filepath

## File System

- Files are stored on storage device. Arrangement of files in storage device is called as "File System".
- e.g. FAT, NTFS, EXT2/3/4, ReiserFS, XFS, HFS, etc.
- File System logically divide partition into 4 sections.
  - Boot block/Boot sector
    - Contains programs/info required for booting of OS
    - Typically contains bootstrap program and bootloader program
  - Super block/Volume control block
    - Contains information of whole partition.
    - Capacity, Label.
    - terminal> df -h
      - Total number of data blocks/inodes.
      - Number of used/free data blocks/inodes.
      - Information of free data blocks/inodes.
  - Inode List/Master file table
    - Inodes (FCB) for each file
  - Data blocks
    - Stores data of the file.
    - Each file have zero or more data blocks.
    - Size of data blocks can be configured while creating file system
- File system is created by the format utility while formatting the partition.

- Windows: format.exe
- Linux: mkfs
  - terminal> sudo mkfs -t ext3 /dev/sdb1
  - terminal> sudo mkfs -t vfat /dev/sdb1
  - -t fs\_type e.g. ext3, ext4, vfat, ntfs, ...
  - partition e.g. /dev/sdb1
- Disk/partition naming conventions
  - Windows:
    - Disks are named as disk0, disk1, ...
    - partitions are named as drives i.e. C:, D:, E:, ...
  - Linux:
    - Disks are named as /dev/sda, /dev/sdb, /dev/sdc, etc.
    - Partitions per disk are named as
      - sda partitions: sda1, sda2, sda3, ...
      - sdb partitions: sdb1, ...

## Linux File Structure

- Linux follows "/" (root) file system.
- "/" is a starting point of Linux file system.
- All your data is stored in this partition.
- / contains boot, bin, sbin, etc, root, home, dev, proc, mnt, media, opt
- In Linux everything is a file.
- Mainly there are two types of files in Linux
  - File
  - Directory (Folder)
- Linux Directories
  - boot - files related to booting
    - vmlinuz - kernel Image
    - grub - boot loader
    - config - kernel configuration
    - initrd/initramfs - initail root file system
  - bin - user commands in binary format
  - sbin - all admin/system commands in binary format
  - etc - configuration files
  - root - home directory of root user
  - home - it contains sub directories for each user with its name
    - devendra -> /home/devendra
    - sunbeam -> /home/sunbeam
    - osboxes -> /home/osboxes
  - dev - it contains all device related files
  - lib - shared program libraries required by kernel
  - mnt - it is temporary mount point
  - media - it is mount point for media eg cdrom
  - opt - stores optional files of large softwares

- proc - virtual file system - it contains information about system or processes
- sys - entries of each block devices, subdirectories for each physical bus type supported, every device class registered with the kernel, global device hierarchy of all devices
- tmp - temporary files that may be lost on system shutdown
- usr - read only directory that stores small programs and files accessible to all users

## Path

- It is a unique location of any file in the file system.
- It is represented by character strings with few delimiters ("/", "\", ":")
- Types of path
  - There are two types of paths in linux
  - Absolute path
    - Path which starts with "/" is called as absolute path.
    - E.g. /home/devendra/MyData/Demos/demo01.sh
  - Relative path
    - Path with respect to current directory is called as relative path
    - E.g. MyData/Assignments/assign02.pdf

## Linux commands

- cd
  - cd ~ - change working directory to home directory
  - cd - - change working directory to old working directory
  - cd .. - change working directory to parent directory
- ls
  - ls - list the contents of present working directory
  - ls path - list the contents of given path
- touch
  - if file does not exist, empty file is created
  - if file exist, timestamp of that file is changed
- pwd
  - pwd - print current working directory
- mkdir
  - mkdir dir - create directory
  - mkdir -p a/b/c - create nested directories
- rmdir

- rmdir dir - remove empty directory
- rm
  - rm file - remove file
  - rm -r dir - remove directory with contents
  - rm -rf dir - force remove directory
- cat
  - cat file - display file content
  - cat > file - create new file
  - cat >> file - append to file
- Special Characters
  - ~ - home directory
  - . - current directory
  - .. - parent directory
  - / - root directory separator
  - \ - escape character
  - \* - matches multiple characters (wildcard)
  - ? - matches single character
  - > - redirect output (overwrite)
  - >> - redirect output (append)
  - < - redirect input